

Guglavath Tharun

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EDUCATION

Indian Institute of Technology, Guwahati

Nov 2021 – May 2025

- B.Tech in Mechanical Engineering

TECHNICAL SKILLS

Programming Languages: C++ , C, SQL, Python, Matlab*

Web Technologies: HTML, CSS, JavaScript*

Database Management: MySQL

Tools/Frameworks: NumPy, Pandas, Tensorflow, scikit-learn*, matplotlib, Gradio, Streamlit, OpenCV*

Miscellaneous: MS Excel, Power BI, Figma, Tableau * Elementary Proficiency

KEY COURSES TAKEN

- Data Structures and Algorithms
- Basic Calculus
- Optimization Methods in Engineering
- Introduction to Computing (Lab + Theory)
- Machine Learning
- Linear Algebra
- Industrial Engineering & Operations Research
- Ordinary Differential Equations

EXPERIENCE

Data Science Intern, Adani Electricity, Mumbai

May 2024 – July 2024

- Advanced Time Series Forecasting:** Developed a **Multivariate Auto Regressive Time Varying (ARTV)** Model using **Statsmodel (ARIMA)** to forecast **Day-Ahead Electricity Demand**, achieving a **MAPE** of **5%** and an **R² Score** of **90%** on the test set.
- Leveraged **Python** and **Pandas** for **robust feature engineering**, performing **Seasonal** and **Cyclical** transformations, capturing complex **demand patterns** driven by weather fluctuations and events.
- Managed a large-scale **dataset of 4 years** (142,976 data points), addressing missing values, COVID-19 data gaps, and refining event/holiday features to **enhance model robustness**.
- Executed model development and scaling on **Google Colab**, optimizing processing for large datasets.

PROJECTS

Medicine Recommendation System

July 2024 – Aug 2024

- Implemented a Machine Learning-based **Recommendation System** using **Support Vector Classifier (SVC)**, selected after comparing multiple models confusion matrices & accuracy scores, **to predict diseases** and **recommend medications, diets, precautions, and descriptions** based on user symptoms.
- Integrated into a dynamic **Gradio** and **Streamlit** interface for **real-time interaction**, allowing users to input symptoms via checkboxes and receive personalized medical recommendations instantly.
- Utilized **scikit-learn**, **Pandas**, and **NumPy** for model development and deployment, training on a dataset of **4940 data points** with **132 symptom features**, along with diet, precaution, medication, and description data, enhancing its application in **healthcare**.

Optimizing the Charging Time of Electric Vehicles

Jan 2024 – May 2024

- Designed and conducted a survey of **500 individuals**, revealing key consumer insights: **60% owned cars**, but only **4% owned EVs**. Identified pain points are high purchase costs, limited driving range, limited charging stations and long charging time.
- Proposed a novel **EV charging network** along major highways in India through partnerships with **Dhaba owners**, enhancing accessibility and convenience of charging stations for long-distance travel, while ensuring safety and achieving a **25% reduction** in initial design costs.

POSITION OF RESPONSIBILITY

Core Team Member, Election Commission Design Team IITG 2024 :

- Contributed to the design and development of digital and print assets for the 2024 election campaign, significantly enhancing voter outreach.
- Collaborated with cross-functional teams to create impactful communication materials, contributing to an **8.4%** increase in votes from the previous year.

ACHIEVEMENTS

- Flipkart Grid 6.0 2024** : Participated and qualified to **Level-2** of Flipkart Grid 6.0 - **Software Development** Track organised by Flipkart.
- Essay Writing on ACB 2016** : Awarded a certificate of Appreciation for securing **2nd** position in the essay writing competition conducted on the eve of **Anti-Corruption Week**.
- Platoon Commander, National Cadet Corps 2018** : Junior Division, 32 (T) Bn Adilabad, 2018.
- Company commander in Training Camp of (NCC) 2017** : Successfully organized & executed various training sessions and drills, ensuring cadets were well-prepared for competitions securing "**A**" certificate.